

Externalizing rescue operations at sea: The migration deal between Italy and Libya

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Abstract

Between 2010 and 2019, the global number of forcibly displaced individuals doubled, prompting countries to sign migration deals. Under these agreements, transit or origin countries halt migrants in exchange for support. The 2017 Italy–Libya deal outsourced search and rescue operations in the Central Mediterranean to the Libyan Coast Guard (LCG), aiming to stop migrants from reaching Italy. Using a spatial difference-in-differences design, we find an increase in the probability of observing a deadly incident in areas patrolled by the LCG post-deal. Descriptive evidence suggests a shift in routes, diverting a significant portion of migration to the Western Mediterranean Route.

Keywords: Mediterranean; migrants; Libya; border control

JEL classifications: F22, H12, K42

1. Introduction

The number of forcibly displaced individuals doubled between 2010 and 2019 (UNHCR 2019), causing a sharp increase in global migration flows. During these years, migration became one of the most frequently discussed topics among policymakers, academic scholars, and the general public. In Europe, migration increased after 2011 and the start of the so-called *Arab Springs*. The number of first time asylum applicants in the European Union (EU) went from little less than 200,000 in 2010 to more than one million in 2015 and 2016. The number decreased to 620,000 in 2017 and plateaued after that. The number of asylum seekers in the EU started growing again in 2022, with little less than 900,000 asylum applications (Eurostat 2023). The number of forcibly displaced migrants is strongly correlated with the number of deaths while migrating, as international migration of forcibly displaced individuals is a risky endeavor. The International Organization for Migration (IOM) estimates that after decreasing since 2017 the number of global deaths linked to migration reached 6,876 people, the highest death toll recorded since 2016 (IOM 2023).

A significant portion of individuals in developed countries view immigration as a threat. Natives often perceive migrants as more numerous, culturally distant, and less qualified than they actually are, which leads to the belief that migrants are likely to be a burden on society.¹ The recent rise in popularity of extreme far-right parties often reflects the strong demand from natives to mitigate migration flows (Halla, Wagner, and Zweimüller 2017). The increasing number of migrants and the strong negative reactions from natives pose a dilemma for policymakers; on the one hand, human rights advocates and international humanitarian organizations push for a strong effort by developed economies

¹ For a comprehensive review of the literature on misperceptions about immigration, see Lutz and Bitschnau (2023).

to save the lives of migrants and facilitate their path toward a new life without poverty and violence—for example, the United Nations High Commissioner for Refugees (UNHCR) recently released a call for decisive action following the death of between 400 and 750 migrants off the coast of Greece on 14 June 2023 (UNHCR 2023). On the other hand, a large fraction of natives often push for restricting migration flows, as they are afraid that migrants will compete with them on the labor market and profit from the social welfare systems of developed countries (Constant 2014; Peri 2014).

Migration deals are an attempt to overcome this policy dilemma, with the aim of helping migrants to escape violence, while also stopping them from reaching their destination countries. *Migration deals* are typically international treaties or agreements between receiving and transit or sending countries, whereby transit countries provide humanitarian assistance to migrants but also stop them from reaching the destination country, in exchange for financial and logistical support by the destination country. Notable examples of *migration deals* include the numerous deals signed by the USA with Mexico to stop migrants from crossing the border (The Guardian (2021a)), as well as the agreement between the EU and Turkey to stop migrants from reaching Greece (International Rescue Committee 2022). More recently the UK has also signed a deal with Rwanda, to deport migrants who reach the country illegally (The Guardian 2022), and a similar deal was signed in November 2023 between Italy and Albania (CNN 2023). In July 2023, the EU signed off on a deal with Tunisia to help stem irregular migration to combat the criminal gangs involved in trafficking and people-smuggling (The Guardian 2023). *Migration deals* have often been effective in deterring migration toward developed countries, however, they have often been criticized as a way for developed countries to shrug off their humanitarian protection responsibilities, in particular when they involve a country with weak institutions (Crawley 2022).²

In this study, we examine the impact of a *migration deal*—the Memorandum of Understanding (MoU)—between Italy, a country with strong institutions, and Libya, a country with relatively weak institutions. The deal was signed in February 2017. The disparities in institutional strength between Italy and Libya are a crucial aspect of the agreement, influencing the incentives of both countries during rescue operations. Libya, in particular, may be motivated to minimize arrivals and discourage departures by providing inadequate assistance to distressed boats. Furthermore, given that Libya's efforts in rescuing migrants depend on the financial and logistical support of Italy, Libya might choose to exert varying levels of effort in rescuing migrants as a strategic tool for blackmail.

Under this international agreement, Italy started delegating border control and search and rescue (SAR) activities to Libyan authorities. The MoU also enhanced cooperation between the two countries to monitor Libya's southern border (Gentiloni and Favez 2017). In May 2017, Italy donated the Libyan Coast Guard (LCG) four patrolling boats (analisiidifesa.it 2017), enabling the LCG to conduct SAR operations.³ In July 2017, the Italian authorities asked all NGOs operating in the Mediterranean to sign a Code of Conduct (Avvenire 2017). This Code of Conduct could potentially create several obstacles to humanitarian action according to several NGOs operating in the Central Mediterranean (The Guardian 2017a). The shift in responsibility for SAR operations completely changed the migrant's expected outcome of being rescued. Before the delegation of SAR activities to Libya, migrants rescued by Italian authorities or by humanitarian NGOs would be brought to Italy, after the delegation of SAR activities to the LCG, migrants rescued by the LCG would be brought back to Libya.

The subsequent years showed a significant decline in the number of migrants crossing the Central Mediterranean route, which supporters attribute to the MoU's role in reducing migration flows. However, critics argue that entrusting border control to the LCG exposed migrants to higher risks of shipwrecks and fatalities in the Mediterranean Sea, along with inhumane conditions in Libyan detention centers (BBC 2019). Despite these concerns, the MoU was renewed in February 2020 without modifications (Avvenire 2020) and automatically renewed in November 2022 (Il Sole 24 Ore 2023).

In this article, we address the following questions: does outsourcing SAR operations to a weak state increase the likelihood of migrant fatalities during migration attempts? Do migrants respond to changes in SAR operations by changing migration routes? By analyzing the specific context of the

² Arrivals in Greece dropped by 80 per cent after the EU signed the *migration deal* with Turkey <https://www.consilium.europa.eu/en/policies/eu-migration-policy/eastern-mediterranean-route/>

³ Libya was in a civil war when the MoU was signed and did not have any technical equipment to provide SAR in the Central Mediterranean. We provide more context in the institutional setting section of the paper, where we provide an overview of the support that Italy granted to Libya.

Italy–Libya MoU, we assess the consequences of outsourcing humanitarian assistance to countries with limited institutional capacity.

The MoU starts the process of externalization of SAR Operations in the Central Mediterranean to the LCG. Before the signing of the MoU, the ships of the Italian Coast Guard and humanitarian NGOs were allowed to operate in the entire Mediterranean. These ships often rescued migrants right in front of the Libyan coasts. After the signing of the MoU, both the Italian Coast Guard and NGOs were gradually forced away from Libyan territorial waters by the LCG, which claimed a monopoly on SAR operations even beyond its territorial waters ([The Guardian 2017c](#)).

To test whether the outsourcing of SAR operations to the LCG made the Central Mediterranean route riskier, we use a spatial difference-in-differences (DiD) research design. We compare the probability of observing deaths or disappearances close to the Libyan coast, where the LCG takes over after February 2017, and close to Italy, where the Italian coast guard is in charge of SAR operations. We control for systematic differences, spatial spillovers, and differential time trajectories for the probability of dying or disappearing across areas of the Central Mediterranean. Moreover, we also control for variations in the traffic of commercial ships across the Mediterranean—to account for potential responses to the policy by commercial vehicles—as well as meteorological conditions. We complement our DiD research design with non-parametric treatment effect estimates for the distance from Tripoli (where the LCG was operating).

Our identification strategy allows us to test if the geographic distribution of shipwrecks changed after the signing of the MoU (i.e. when the LCG was supposed to provide SAR operations). Under the assumption that the exact location of a shipwreck occurs at random—after accounting for factors such as the roughness of the sea, the specific risk factor of a specific area of the sea, etc.—the spatial distribution of shipwrecks should not change after the signing of the MoU. In other words, if SAR operations are conducted effectively both by the LCG and by the Italian Coast Guard, we should still observe shipwrecks after the signing of the treaty, but their geographic distribution should not change.

Our results provide causal evidence that the areas where the LCG was more likely to provide SAR operations experienced an increase in the probability of observing a deadly event. We estimate that, on average, the probability of observing a shipwreck, after the passage of the MoU decreases by 50 per cent for every 100 km of distance from the Libyan shores. Moreover, our non-parametric treatment effect estimation reveals that our results are driven by the areas of the Central Mediterranean close to Tripoli—where the LCG operated—with the probability of observing a deadly event increasing eight times in areas that are within 120 km from Tripoli. Our dynamic treatment effect estimates show that the effect is concentrated during the months of May and June of 2017. This coincides with the donation of the four patrolling boats that enabled the LCG to conduct SAR operations ([analisi difesa.it 2017](#)). The dynamic treatment effect estimates also show that we estimate no treatment effect starting from July 2017. This coincides with three major events: the Italian Government's NGOs code of conduct, the beginning of monitoring activities by the EU authorities—described in more detail in the institutional context section—and a sharp drop in departures.⁴

We provide suggestive evidence for a strong diversion effect toward the Western Mediterranean route, where migrants reach Europe through Spain, mainly after passing through Algeria. The number of arrivals in Italy dropped sharply after the probability of observing a deadly event in areas patrolled by the LCG increased—going from a peak of over 27,000 arrivals per month in October 2016 to little less than 6,000 arrivals in October 2016. Arrivals in Spain doubled in the same period, going from little more than 2,000 arrivals in October 2016 to more than 4,000 in October 2017. We show that after the passage of the *Memorandum*, the ethnic mix of migrants arriving in Spain sharply changed. In a DiD framework, we compare changes in the arrivals of individuals from countries that used to migrate to Italy prior to the MoU with changes in the arrivals of individuals from countries that were not likely to migrate to Italy prior to the MoU. We find that after the signing of the MoU, the number of migrants arriving in Spain and that come from countries that would normally arrive in Europe through Italy increased by roughly 40 per cent.

This article contributes to several strands of the literature. First, it expands on the literature about illegal migration deterrence through tighter border control and sanctions ([Hanson and Spilimbergo 1999](#); [Angelucci 2012](#); [Bazzi et al., 2018](#); [Feigenberg 2020](#)). This literature generally finds that tighter border control is effective in reducing migration attempts, thus being in line with our findings. We

⁴ We document the sharp drop in departures in the [Supplementary Appendix Subsection A.4](#).

contribute to this literature by examining *migration deals* as an additional policy tool for deterring illegal migration through tougher border control.

Second, we contribute to the literature about the behavioral responses of migrants to tougher border control policies. [Cornelius \(2001\)](#) and [Gathmann \(2008\)](#) show that tighter border control in Mexico led migrants and smugglers to opt for more expensive and riskier routes, in order to avoid border enforcement. These results are consistent with our findings about the displacement effect of the *Memorandum*. In a broader way, our article also relates to the literature that tries to model the decisions of illegal migrants and smugglers ([Friebel and Guriev 2006](#); [Woodland and Yoshida 2006](#); [Tamura 2010](#); [Friebel et al., 2024](#)).

Our article is strongly related to the emerging literature on the situation in the Central Mediterranean after the major historical events in Africa and the Middle East after 2011. [Deiana, Maheshri, and Mastrobuoni \(2024\)](#) analyze the behavioral response of human smugglers to SAR operations, finding that smugglers switch to unsafer and cheaper boats when SAR operations intensify. [Battiston \(2022\)](#) shows the existence of a dynamic tradeoff between present rescue policy and migrants' departures, as decisions by the policymaker are going to affect migrants' expectations. The closest article to ours in the literature is [Fasani, Frattini, and Weisser \(2019\)](#), the authors find a relationship between SAR operations from 2012 to 2017 and the probability of dying in the Central Mediterranean. Our article differs from [Fasani, Frattini, and Weisser \(2019\)](#), as we are considering outsourcing SAR operations. We explicitly consider the effect of the MoU and the subsequent externalization, a policy that was implemented after the period of analysis by [Fasani, Frattini, and Weisser \(2019\)](#). Our article contributes to this literature by providing the first estimates of the impact of externalizing rescue operations in a foreign country with weak institutions. Moreover, we add to this work by providing the first systematic evidence for the displacement effect of the *Memorandum* on other migration routes.

2. Institutional setting

The area of the Mediterranean between Italy and Libya has been the stage of important political and social developments during the last decade. Following the fall of Qaddafi's regime in 2011 and the start of the civil conflict in Libya, migration flows through the Central Mediterranean have risen dramatically, going from less than 10,000 crossings in 2010 to more than 170,000 crossings in 2014. Consequently, Italy and the EU invested huge amounts of resources to address humanitarian and border control needs. In what follows, we provide an overview of migration dynamics in the Central Mediterranean and a timeline of the main policy changes that occurred, along with a thorough description of the MoU signed by Italy and Libya.

2.1 The Central Mediterranean route

The Central Mediterranean has become the deadliest border in the world ([Reuters 2017](#)). Between 2013 and 2020, 680,000 migrants reached the Italian coasts and more than 17,000 died while attempting to cross. Tragic events, such as the large shipwreck of October 2013 that killed more than 360 migrants, led Italian and European authorities to conduct SAR Operations in order to make the route safer ([Deiana, Maheshri, and Mastrobuoni 2024](#)). Such operations mainly include patrolling the Mediterranean and rescuing migrants' boats in distress.

The Central Mediterranean route mainly concerns Italy and Libya, 90 per cent of departures are from Libya and almost all arrivals are in Italy ([IOM 2015](#)). Geographic proximity with Italy and political instability in Libya are the major factors that brought smugglers to choose Libya as their favorite departure point. In 2014, after the Civil War in Libya, 170,000 immigrants landed in Italy, compared to only a little more than 40,000 during the previous year.

Alternative routes to the Libyan–Italian border crossing experienced tighter monitoring and tougher enforcement of anti-migration policies. Since 1998, human trafficking from Tunisia—Italy's closest African country—has been heavily reduced, thanks to an agreement between the Italian and the Tunisian governments ([The Irish Times 1998](#)). Over three years, Italy paid nearly 80 million euros to Tunisia in order to enforce border control and build detention centers for illegal immigrants. Moreover, Italy agreed to set entry quotas for Tunisian citizens every year. The only notable inflows from Tunisia to Italy happened in 2011, after the uprisings that sparked the *Arab Spring*. The signing of a further agreement between Italy and Tunisia reduced the flows to pre-2011 level ([Reuters 2011](#)).

2.2 SAR operations in the Mediterranean

SAR operations in the Mediterranean have experienced significant variations in scale and declared objectives. In general, operations led by the EU mainly aimed at securing the external borders of the EU, whereas operations by the Italian Government and NGOs explicitly had the objective to make the route less risky. The strategy adopted by the Italian government and by NGOs has often been accused to exercise a *pull-factor* for illegal crossings, by making the Central Mediterranean safer (Washington Post 2014). In Figure 1 we provide a timeline of the the main operations that took place in the Central Mediterranean between 2014 and 2018. We provide a detailed description of the main operations that were held in the Central Mediterranean in Subsection A.1 in the Supplementary Appendix.

2.3 The externalization of SAR activities to Libya

To address the challenges of migration to Italy and mitigate the perceived *pull factor* associated with SAR operations led by the Italian Coast Guard and several NGOs, Italy embarked on a collaboration with the Libyan Government of National Accord (GNA). The primary objective of this joint effort was to transfer the coordination of SAR activities in the Central Mediterranean to Libya (Human Rights Watch 2019).

By shifting the coordination to Libya, the intended outcome was to alter the standard procedure of rescuing distressed boats in the region. Prior to 2017, rescued migrants were typically brought to Italy. However, with this new arrangement, the rescued individuals would be brought back to Libya instead.

At the time when the collaboration between the Italian Government and the GNA started, Libya was in the middle of a civil war. Libya has been in a civil war since the start of 2014 when the General National Congress (GNC) refused to hold new elections at the end of its political mandate. Many actors were involved in the conflict (including many militia groups) and, at the end of 2019, deaths were estimated to be around 11,000 (University 2022).

The start of 2014 saw the division of Libya under two areas of influence: one controlled by the GNC—based in Tripoli—and another ruled by the new House of Representatives from Tobruk. The first ceasefire was agreed upon at the end of 2015 when the GNA was formed after a peace deal backed by the UN.

In this context, Italy signed the MoU with the Libyan GNA. At the start of 2017, Italy started the process of outsourcing border control activities and SAR operations to the LCG, with the intent to “achieve solutions regarding some matters that negatively affect the Parties, including the irregular immigration phenomenon” (Gentiloni and Fayez 2017). The agreement was endorsed by the EU at its summit in Malta in February 2017 (Baldwin-Edwards and Lutterbeck 2019).

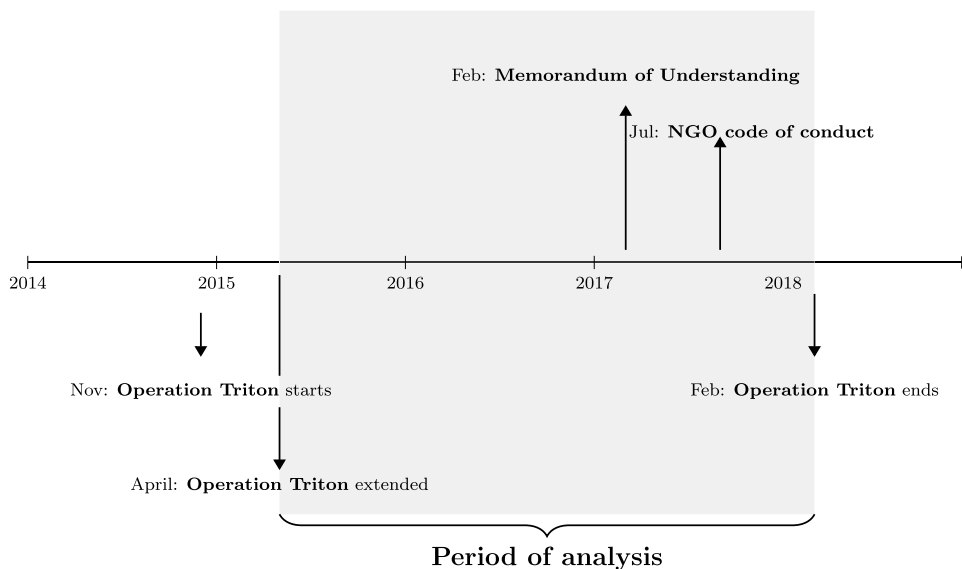


Figure 1. Timeline of border control policies in the Central Mediterranean, 2014–19.

The transfer of SAR activities to LCG took place gradually, starting in February 2017. At the time the MoU was signed, the GNA lacked the necessary equipment and organizational capacity to conduct SAR operations. Specifically, they did not have enough boats for SAR activities, and they were also without a Maritime Rescue Coordination Center (MRCC) (Human Rights Watch 2019).

To address these limitations, the Italian government stepped in to assist. They agreed to provide funding and training to boost the capabilities of the LCG in the Tripoli area, which is a significant departure point for migrant ships. Back in 2017, this Coast Guard unit was severely underequipped and understaffed (The Guardian 2017b). The collaboration aimed to enhance the ability of the LCG to conduct SAR operations and manage migration-related challenges in the region.

In May 2017, Italy provided the LCG with four fast patrol boats (analisiidifesa.it 2017). Around the same time, the Italian MRCC began sharing information about boats in distress with the Libyan authorities (Human Rights Watch 2019). Additionally, rescue NGOs were instructed to wait before conducting SAR activities to allow the LCG to intervene effectively. Reports from NGOs operating in the Central Mediterranean raised concerns about the LCG's performance during SAR operations. There were accounts of the LCG approaching migrant boats and firing shots into the air, which allegedly caused panic among the migrants on board. As a result of the shots being fired, some panicked individuals jumped into the sea (Amnesty International 2017). This incident raised questions about the safety and effectiveness of the LCG's approach to managing distress situations at sea.

In July 2017, the Italian authorities asked all NGOs that were active in the Mediterranean to sign a Code of Conduct, which had the potential to create several obstacles to humanitarian action.⁵ While not all NGOs agreed to back the new provisions, only those who did were kept in the loop of the internationally organized rescue system, managed by the Maritime Rescue Coordination Centre in Rome.

In July 2017, the EU Naval Force Mediterranean (ENFM), was tasked by the EU to monitor the LCG and the Libyan Navy. This monitoring included periodic meetings between the ENFM and the LCG and the Libyan Navy. The first report released by the ENFM (ENFM 2018) highlights how the training that started in July 2017 has been successful in making the LCG and the Libyan Navy more effective. The report also highlights, however, how the initial conditions in which the LCG was operating were critical, with limited communication systems, power supply, telephones, and personal computers. The monitoring of the operations of the LCG was then improved by providing the LCG and the Libyan Navy with Go-Pro cameras, in order to record activity in real time.

The externalization of SAR activities to Libya, which started with the MoU, is similar to other agreements signed with transit countries for refugees, such as the 1998 agreement between Italy and Tunisia and the 2016 agreement between the EU and Turkey. However, the context in which the Memorandum was signed was different with respect to the 1998 and 2016 agreements, as Libya was the biggest departure hub for migrants at the time. Finally, the Memorandum was signed between Italy and the GNA, which represented only one of the factions that was fighting the Libyan Civil War. This setting offers therefore a unique opportunity to study the effects of externalization of SAR activities in a setting where institutions are extremely weak.

3. Data and descriptive evidence

In this article, we use data from several sources. In particular, we combine incident-level data on deaths that occurred while migrating with detailed information about weather conditions, wave height, the intensity of SAR operations conducted by NGOs, and proxies for maritime traffic in the Mediterranean. We also rely on data about migration intentions collected at migration hot spots in Niger.

3.1 Deaths in the Mediterranean

The information on deaths and people going missing in the Mediterranean comes from the *Missing Migrants* project, managed by the IOM. The IOM collects data on individuals who died in the process of

⁵ For example, the international humanitarian organization *médecin sans frontière* claimed that the Code of Conduct would impose unreasonable restrictions while conducting search and rescue operations (Camilli 2017)

migrating toward an international destination, regardless of their legal status. For the Mediterranean, the sources of information are the relevant national authorities: mainly national and European Coast Guard organizations. The reliability of the data is further validated through exchanges between the IOM and UNHCR.

The Missing Migrants data only report deaths and disappearances that occurred while migrating. Therefore, our dataset excludes individuals who died in detention facilities or after repatriation, as well as during labor exploitation in foreign countries. Deaths that occurred in refugee camps or after migrants were established in a new country are also not counted, as are deaths of internally displaced persons.

The statistical unit of our data is a migration incident, for which we observe the geographical coordinates, date, the number of deaths and a minimum estimate of missing people, the number of survivors, the source of the information (alongside an estimate of source quality), the migration route and the likely cause of death. For our main specification, we restrict our analysis to drownings that occurred on the Central Mediterranean route, as they are the most likely reported cause of death in case of a shipwreck.

As the IOM reports, collecting data on deaths linked to migration is often challenging. The irregular nature of most migration routes, the remote locations where many incidents occur and the involvement of criminal actors are likely to give rise to issues of underreporting. In the [Supplementary Appendix Subsection A.3](#), we discuss in detail all the potential limitations of using these data and under what conditions our estimates could be biased.

[Figure 2](#) plots all deadly incidents in the Central Mediterranean from 2014 to 2019. Deaths after the signing of the *Memorandum* (red dots) tend to be more concentrated close to the Libyan coast, than deaths before the signing of the *Memorandum* (gray dots). This distribution of deaths before and after the signing of the *Memorandum* is consistent with a poorer quality of SAR operations by the LCG.

3.2 Additional data

Throughout the article, we use data about Arrivals in Europe and estimated departures from Libya. We use these data in model specifications where we include a rich set of control variables as well as to document the potential diversion effects of the externalization of SAR activities. We also use data on migration flows in North Africa, sourced by the Flow Monitoring Program of the IOM. We use these data to document the change in migration patterns within Africa after the MoU. Finally, we complement the information about deaths, arrivals, and migration flows with data about NGO's SAR activities, meteorological, and sea conditions, the type of boats that migrants use while attempting to cross the Mediterranean, as well as data about conflict in Africa, in order to account for potential confounding factors. We give a more detailed description of the data described above in the [Supplementary Appendix Subsection A.2](#).

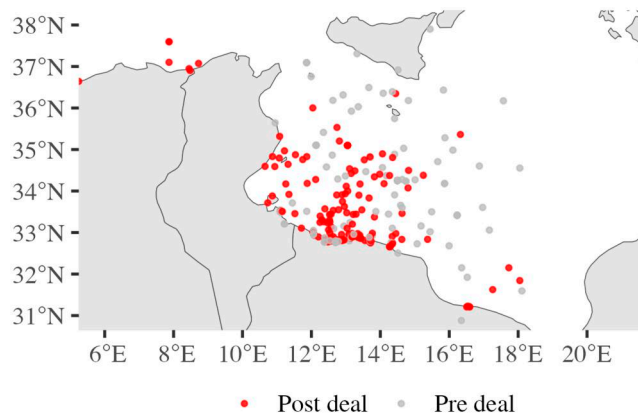


Figure 2. Deaths in the Central Mediterranean, 2014–19.

Notes: This map plots the location of deaths in the Mediterranean from 2014 to mid-2019. Gray dots indicate deaths that happened before the MoU, red dots indicate deaths occurred afterwards.

4. Methodology

To assess the causal impact of externalization of SAR activities on the probability of observing a deadly event, one would like to randomly assign the location of SAR operations to the Italian and the LCG. By comparing the probability of observing a deadly event in areas patrolled by the Libyan and the Italian coast guard one would be able to obtain the causal impact of externalizing SAR operations to the LCG on the probability of observing a deadly event while migrating. However, such a thought experiment is not feasible, as SAR operations areas are agreed upon by governments and international organizations and, for obvious reasons, are not assigned randomly.

We overcome this issue by using a DiD approach. We compare the change in the probability of observing deadly events in areas closer and further away from the Libyan coasts before and after the signing of the Memorandum. The underlying logic of this approach is that we expect areas that are closer to the Libyan coast to be more likely to be patrolled by the LCG after the signing of the Memorandum in January 2017. We limit our analysis to the period between the extension and the end of Operation Triton (April 2015 to February 2018). By using this time frame, we avoid capturing the effect of other policy changes that affected the risk of dying while migrating through the Central Mediterranean route.

We first divide the central Mediterranean into ninety-four quadrants of one degree of latitude and longitude. For each quadrant, we compute the minimum distance from the Libyan coast, which represents our continuous treatment measure. We then estimate the following DiD linear probability model:

$$y_{qt} = \delta[\text{distance}_q \times \text{Post}_t] + \beta[\text{distance}_q \times X_t] + \gamma SC_{qt} + \lambda_t + \lambda_q + \varepsilon_{qt}, \quad (1)$$

where y_{qt} is a dummy variable that takes value one if there was at least one deadly event in quadrant q in month t . The dummy Post_t takes the value 1 if month t is after the Memorandum.

The vector X_t includes a set of relevant time-varying controls, such as the number of NGOs active in the Mediterranean the number of conflict-related events in Libya, and the number of ships crossing the Suez Canal as a proxy for maritime traffic in the Mediterranean. We interact this set of covariates with our measure of distance, thereby modeling the differential impact on the probability of dying that they can have across different areas of the Mediterranean. In some specifications, we interact the set of time-varying covariates with quadrant fixed effect. This approach allows us to account for any non-linearities in the impact of the number of NGOs, or conflict-related events in Libya, and of marine traffic on the probability of dying across different areas of the Mediterranean. In some specifications, we augment the model by controlling for estimated departures from Libya interacted with quadrant fixed effects.⁶ Moreover, we also control for average sea conditions during month t in quadrant q (SC_{qt}).

We include a vector of month fixed effects λ_t that allows us to absorb any common shock to the probability of dying across quadrants, for example, the average quality of boats departing in a certain month. We also include quadrant fixed effects λ_q , allowing us to exploit within-quadrant variation in the probability of observing a deadly event and therefore accounting for systematic differences in the probability of observing a deadly event between quadrants. The stochastic error term ε_{qt} is clustered at the quadrant level.

In some specifications, we control for quadrant-specific time trends by including a term $\lambda_q \times \text{Time}_t$. Time trends allow us to isolate the effect of the Memorandum from quadrant-specific dynamics starting before January 2017.

Furthermore, in some specifications, we account for spatial dynamics through an additional term, W_{jt} . The matrix W contains the spatial weights associated to each possible quadrant pair (i, j) . Weights are computed as $w_{ij} = \frac{1}{e^{\text{distance}_{ij}}}$ if $i \neq j$ and 0 otherwise. The vector y_t represents the outcomes in all quadrants in month t . The inclusion of spatial dynamics is important for several reasons: on one hand, deadly events might cluster in particular areas because of sea or weather conditions; on the other, the probability of deadly events in quadrant q depends on the realization of events in quadrants that are closer to Libyan coasts. To allow spatial dynamics to change over time, we interact W_{jt} with a set of month dummies λ_t . We extend this approach to control for the impact of changes in sea conditions on neighboring quadrants, in particular, we multiply W by a vector of sea conditions s_t . To allow for the

⁶ We do not control for departures in our main specification as they could potentially be affected by the MoU itself

impact of sea conditions in other quadrants to change over time, we interact W_{st} with a set of month dummies λ_t . This approach allows us to model the impact of sea conditions in neighboring quadrants for each quadrant.

The identifying assumption for interpreting the results of our DD is that, in the absence of the MoU, the probability of a shipwreck occurring close and far away from Libya would have followed similar trends over time. Under this assumption, we can interpret the coefficient of interest δ as the causal impact of externalizing SAR operations to the LCG on the probability of observing a deadly event closer to the Libyan coasts, where the LCG is supposed to conduct SAR activities. In particular, the coefficient estimates the change in the marginal probability of observing a deadly event when the distance from Libya increases by 100 km.

We estimate dynamic treatment effects with an event-study design. This allows us to provide evidence for parallel trends in the pre-treatment period and to show how the effect of externalizing SAR operations evolved over time. To do this we estimate the following equation:

$$y_{qt} = \sum_{t=Jan,2015}^{t=Feb,2018} \gamma_d(\text{distance}_q \times \text{Month}_t) + \beta[\text{distance}_q \times X_t] + \gamma SC_{qt} + \lambda_t + \lambda_q + \varepsilon_{qt}, \quad (2)$$

whereby the set of coefficients γ_d represents an estimate of the dynamic treatment effect, λ_t represents a set of month fixed effects, and λ_q a set of quadrant fixed effects.

In the robustness checks section of the article, we show that our results are robust to using quadrants of different dimensions as our units of observation.

4.1 Impact on the number of deaths

We also test whether the number of deaths far away and close to the Libyan shores changed after the introduction of the memorandum. In particular, we use the inverse hyperbolic sine transformation (Bellemare and Wichman 2020) of the number of deaths occurring in quadrant q at time t . We estimate the same models as we do for Equation (1). The results of this exercise are shown in the Subsection A.5 in the Supplementary Appendix.

4.2 Non-parametric distance treatment effects

We complement our main identification strategy with an alternative estimator proposed by Butts (2023). This method allows us to estimate the treatment effect curve as a function of distance, exploiting the partitioning-based least squares approach developed by Cattaneo, Farrell, and Feng (2020). Intuitively, the estimator partitions the space between Libya and Italy into rings centered in Tripoli—where the LCG is operating. The size of the rings (the bins of distance from Tripoli) is automatically chosen using Cattaneo, Farrell, and Feng (2020). The estimator by Butts (2023) then compares the probability of observing a deadly event for each ring before and after the passage of the MoU.

This method requires that treatment effects become zero somewhere between the distance of 0 and the outer ring without the need to specify the exact distance at which it does. If, after some distance, treatment effects become centered at zero, this suggests that common trends hold, akin to the pre-trends test in event study regressions.

We implement this methodology using a finer grid of quadrants of 0.1×0.1 degrees of longitude and latitude to allow for a smoother estimation of the treatment effect curve, and we use the distance from Tripoli as our treatment variable, to account for the fact that the LCG was operating from Tripoli.

While this estimator does not allow us to estimate dynamic treatment effects, it has the advantage of easing the interpretation of our standard DD estimates by explicitly showing where, along the Central Mediterranean Route, the probability of observing a deadly event changes (Butts 2023).

5. Results

5.1 DiD estimates

Table 1 shows our estimates for the effect of externalizing SAR operations to the LCG on the probability of observing a deadly event far away from the Libyan coast. These estimates show that externalizing SAR operations to the LCG made areas closer to the Libyan coast substantially riskier. Our estimates suggest that the probability of observing a deadly event strongly decreases with distance from the

Table 1. Probability of monthly deaths—distance.

	<i>Dependent variable:</i>						
	Prob. of monthly deaths (Mean = 0.02)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Distance × Post Deal	−0.010** (0.005)	−0.010** (0.005)	−0.014** (0.006)	−0.017** (0.007)	−0.013** (0.006)	−0.013** (0.006)	−0.024*** (0.009)
Log(Conflict events) × Distance			Yes	Yes	Yes		
NGO boats × Distance			Yes	Yes	Yes		
Log(Ships Crossing Suez) × Distance			Yes	Yes	Yes		
Log(Conflict events) × Quadrant FE						Yes	Yes
NGO boats × Quadrant FE						Yes	Yes
Log(Ships Crossing Suez) × Quadrant FE						Yes	Yes
Estimated Number of Departures × Quadrant FE							Yes
Sea conditions			Yes	Yes	Yes	Yes	Yes
Quadrant FE		Yes	Yes	Yes	Yes	Yes	Yes
Date FE		Yes	Yes	Yes	Yes	Yes	Yes
Quadrant time trends				Yes			
Distance-weighted number of deaths × Date dummies					Yes	Yes	Yes
Distance-weighted sea conditions × Date dummies					Yes	Yes	Yes
Fraction of Inflatables × Distance			Yes	Yes	Yes	Yes	Yes
Observations	3,382	3,382	3,382	3,382	3,382	3,382	2,136
Adjusted R ²	0.022	0.101	0.102	0.128	0.124	0.143	0.180

The table shows the outcome of several DD regressions in which the outcome is a dummy for whether we observe any deaths (or missing persons) in quadrant q in month t . The regressor of interest is the interaction between the quadrant's distance from Italy and a dummy for months after January 2017. Standard errors are clustered by quadrant. P-values: * $P < .1$; ** $P < .05$; *** $P < .01$.

Libyan coast. In particular, being 100 km farther away decreases the probability of observing a deadly event by 1–2.3 percentage points, representing a 50 per cent to 100 per cent decrease with respect to the average probability of observing a deadly event in the Central Mediterranean.

Our estimates are stable across different model specifications. Comparing Columns (1) and (2), we can see that estimates are identical with and without the inclusion of quadrant and month fixed effects. In Columns (3)–(7), we show that our estimates are robust to a battery of model specifications. Column (3) shows that our results do not change when we control for conflict intensity in Libya, the presence of NGOs, the number of ships crossing the central Mediterranean, a quadrant-specific measure of the roughness of sea conditions, and an interaction of distance and the fraction of migrants departing from rubber boats, which are considered to be less safe.⁷ Column (4) shows that we obtain similar results when we control for quadrant-specific linear trends. Column (5) shows that our estimate does not change when we control non-parametrically for the probability of observing a death in closer quadrants, our measure of spatial dynamics. Column (6) shows that our results are robust to the inclusion of time-varying controls interacted with quadrant fixed effects, that allow us to control non-parametrically for the conflict intensity in Libya, the presence of NGOs, and the number of ships crossing the central Mediterranean. Finally, column (7) shows that our estimates are robust to the inclusion of a control for the total number of estimated departures from Libya interacted with quadrant fixed effects. When we include a control for the total number of departures our estimates become larger. This result could be due to the fact that the number of departures could potentially be affected by the MoU itself and therefore be a bad control. Moreover, data on estimated departures are only available from 2016. For these reasons, we do not consider the model estimated in column (7) as our main specification.⁸

⁷ The addition of the share of rubber boats used by migrants while attempting to cross is controlling for the fact that Libyan smugglers may decide to increase the number of departures as quickly as possible, using any type of boat available, prior to the passage of the MoU.

⁸ In the [Supplementary Appendix Section A.5](#), we use alternative approaches for controlling for departures, using migrant's migration intentions in Niger as a proxy, as well as the lag of total departures.

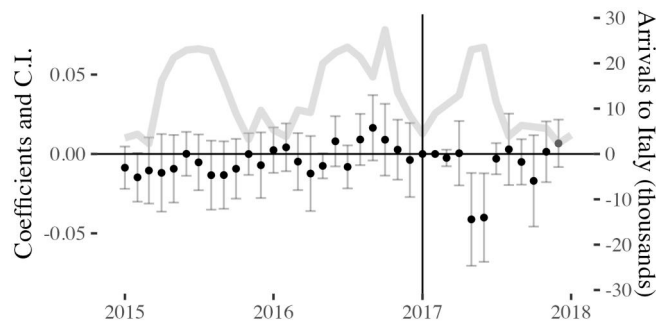


Figure 3. Flexible DD estimation—probability of deadly events.

Notes: The figure displays the results of Equation (2). The coefficients estimate the effect of the interaction of quadrant's distance from Libya (in hundreds of kilometres) and month dummies, along with the associated 95 per cent confidence intervals. The gray line plots the evolution of migrants' arrivals to Italy through the Mediterranean.

Figure 3 displays the results of a dynamic DiD analysis based on Equation (2). The coefficients are estimates of the dynamic effect of a quadrant's distance from Libya on the probability of observing deadly events in that quadrant. Before January 2017, no difference in the probability of a deadly event is observed between areas closer and farther away from Libya. These results support our parallel trends assumption and are consistent with the idea that European authorities provided SAR activities across the entire Mediterranean before the Memorandum. In May and June 2017, 3 months after the Memorandum, we observe a stark difference in the probability of observing a deadly event closer to the Libyan shores. The probability of observing a deadly event after May 2017 goes back to pre-MoU levels in July 2017, coinciding with: the Italian Government's NGOs code of conduct, the beginning of monitoring activities by the EU authorities, and a sharp drop in departures. This result is consistent with the findings in Deiana, Maheshri, and Mastrobuoni (2024), which show how migrants switch to safer vessels after the implementation of the Italian Government's NGOs code of conduct code of conduct.

These results are consistent with the LCG managing SAR operations along the Libyan shores after the signing of the Memorandum. The months in which the effect is concentrated, are May and June 2017, coinciding with a donation of four patrol boats to the LCG by the Italian Government. The official ceremony for this donation ([analisidifesa.it](https://www.analisidifesa.it) 2017) happened at the end of April, and the Italian Government promised to keep supporting the LCG with boats, men, and training in the following months. Our interpretation of these results is that after the LCG actually started managing SAR operations, it crucially failed to provide them effectively, increasing the probability of observing a deadly event in areas that they were supposed to patrol. Several pieces of anecdotal evidence suggest that the personnel of the LCG was not trained in performing SAR operations, and as of 2021, the LCG has been accused multiple times of threatening migrants and firing shots at their boats (The Guardian 2021b).

5.2 Non-parametric distance treatment effects

Figure 4 shows the non-parametric estimates of the differences in the probability of observing a deadly event after the passage of the MoU for different distance bins. The y-axis is re-scaled with respect to the sample mean of the probability of observing a deadly event in a given quadrant—a value of one indicates an estimated treatment effect equal to the average probability of observing a deadly event in a given quadrant. Horizontal red lines indicate the estimated treatment effect for a given bin and gray areas indicate 95 per cent confidence intervals.

We estimate that for areas that are between 16 and 126 km from Tripoli the probability of observing a deadly event increases by 6.3 times with respect to the sample mean. The effect rapidly decays for areas that are further away from Tripoli and becomes non-statistically different from 0 at the 95 percent confidence level at 174 km from Tripoli. The estimated effects then stabilize around zero at 200 km from Tripoli.

These estimates are consistent with the Libyan coastguard providing poor SAR operations beyond Libyan territorial waters after the signing of the MoU. These estimates show that our results are

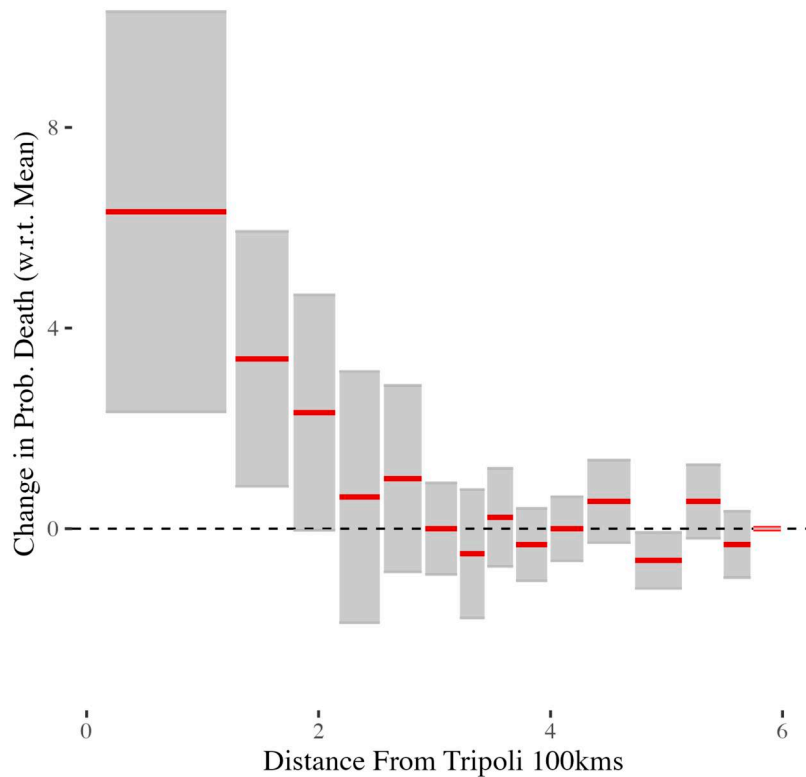


Figure 4. Probability of deadly events.

Notes: The figure provides non-parametric treatment effects estimated using [Butts \(2023\)](#).

unlikely to be driven by a generalized increase in the risk of dying across all areas in the Central Mediterranean.

5.3 Robustness checks

In the [Supplementary Appendix Subsection A.5](#), we show that our results are not driven by arbitrary choices made when setting up our identification strategy. In particular, we show that our findings are robust to alternative definitions of quadrants of different dimensions and to a continuous version of our outcome variable. Furthermore, we provide evidence that the data about migrant's deaths are unlikely to suffer from human error in geo-coding the data about deaths in the Mediterranean. We also provide a theoretical argument to show that our estimates are unlikely to be affected by potential spillover effects that we are not able to capture in our model. Acknowledging that our control for departures might induce *bad control* bias, we show that our results are unaffected by choosing alternative controls for departures. Moreover, we also estimate models, where we weigh our outcome variable by the total number of departures, to explicitly model the possibility that departures could be affected by the externalization of SAR activities to the LCG. Finally, we provide evidence that our results are not mechanically related to the number of departures. We exploit the fact that departures are seasonal also before 2017 and estimate the same model as in our main specification, but using a fake *placebo policy* in February 2016, which does not show any statistically significant effect.

5.4 Migration diversion effects

The externalization of SAR activities to the LCG, which started with the MoU was accompanied by a series of measures aimed at securing Libya's southern border. Italy would be responsible for helping Libya reach this objective, through military assistance and training of the Libyan forces. A tightening of the Libyan southern border was needed, as migrants would not be able to leave the country via sea, to avoid a large stock of migrants being bottlenecked in Libya.

In this section, we show a series of stylized facts that show the following: a sharp switch in migration intentions of transit migrants in Niger, who decide to continue their journey through Algeria instead of Libya, a drastic change in the ethnic composition of migrants arriving to Spain through the Western Mediterranean route, driven by an increase in migrants from nationalities that were previously more likely to migrate to Italy from the Central Mediterranean route. We also show that the probability of observing deadly incidents in the Western Mediterranean rapidly grew after the externalization of SAR activities to the LCG.

These stylized facts are consistent with a change in the behavior of migrants. In particular, these trends suggest that migrants started migrating to Spain opting for routes avoiding Libya. However, it is crucial to interpret these trends cautiously, as they may be influenced by external shocks to migration push factors that we cannot control for, and it is difficult to find an appropriate counterfactual.

We overcome the lack of a counterfactual and test the diversion effect of the MoU using a DiD design exploiting variation in the ethnic composition of arrivals in Spain. The treated units are migrants from nationalities that were likely to migrate to Italy prior to the externalization, and the control group will be composed of migrants from nationalities that were relatively less likely to migrate to Italy prior to the externalization. This setting allows us to partially relax the assumption about external shocks to migration push factors, as we will need push factors to be uncorrelated with the propensity to migrate to Italy before the externalization.

5.5 Descriptive evidence on change in migration decisions

In Fig. 5a, we report the migration intentions of migrants who are surveyed in Niger and intend to go to Libya or Algeria as alternative routes to Europe. Over the 2016–19 period, an average of 61,000 individuals per quarter intended to go to Libya, while approximately 7,000 considered Algeria. Before the externalization of SAR activities, the average for Libya was 98,000, compared to 5,000 for Algeria. After the first quarter of 2017, Libya's average dropped to 11,000, while Algeria's rose to around 9,000, reflecting a strong shift in migration intentions.⁹

In Fig. 5b, we show that the sharp drop in arrivals to Italy is associated with a sharp increase in arrivals to Spain. Figure 5b shows that arrivals in Italy and Spain diverged in mid-2017. When arrivals to Italy start falling, migration flows to Spain begin to increase sharply. While the absolute number of arrivals in Spain remains much lower than in Italy, the raw data suggest that the closure of the Central Mediterranean route diverted some migrants to the closest available route to Europe. Prior to February 2017, Italy averaged approximately 16,000 monthly arrivals, while Spain averaged around 1,000. After February 2017, Italy's average monthly arrivals decreased by approximately 44 per cent, settling at about 9,000. In contrast, Spain witnessed a substantial increase, with arrivals surging by approximately 40 per cent above the pre-February 2017 mean, reaching an average of roughly 1,400. These percentage changes underscore the dynamic nature of migration patterns and the adaptability of migrants in response to evolving circumstances.

Figure 5c plots deadly incidents that occurred in the Western Mediterranean. Gray dots report death incidents that occurred before the passage of the MoU, while red dots indicate death incidents that occurred after the passage of the MoU. It is clear how the number of deadly events in the Western Mediterranean sharply increased after the signing of the MoU. In particular, considering only our sample period, the average number of death incidents went from 2.24 average death incidents per month to 5.92 death incidents per month. The number of deaths went from an average of 14.5 deaths per month to 21.6 deaths per month, increasing by almost 50 per cent.

5.6 Quantifying the diversion effect

To assess whether migrants arriving in Spain would have arrived in Italy in the absence of the MoU, we look at the evolution of the nationality mix of arrivals in Spain over time. We first look at the nationality mix of arrivals in Italy, prior to the signing of the Memorandum, to establish from which countries migrants that were arriving in Italy were coming. We then assign a dummy variable to

⁹ Flows to Libya and Algeria both fall heavily starting in mid-2016, when Niger scales up the implementation of a law designed to fight irregular migration (Law No. 2015-36). The implementation of the law aimed at creating an imaginary red line that migrants were not allowed to cross. The imaginary red line included all areas to the northeast of Agadez (Dauchy et al., 2022), therefore, including the key transit points to go to Algeria and Libya. In particular, our data contain migration intentions when migrants are in the city of Agadez and we use answers on whether they want to reach Arlit (closer to Algeria) or Seguedine (closer to Libya). However, most reports refer to a stronger enforcement of the law in the territory between Agadez and Seguedine, in the direction toward Libya (Tchilouta et al., 2023).

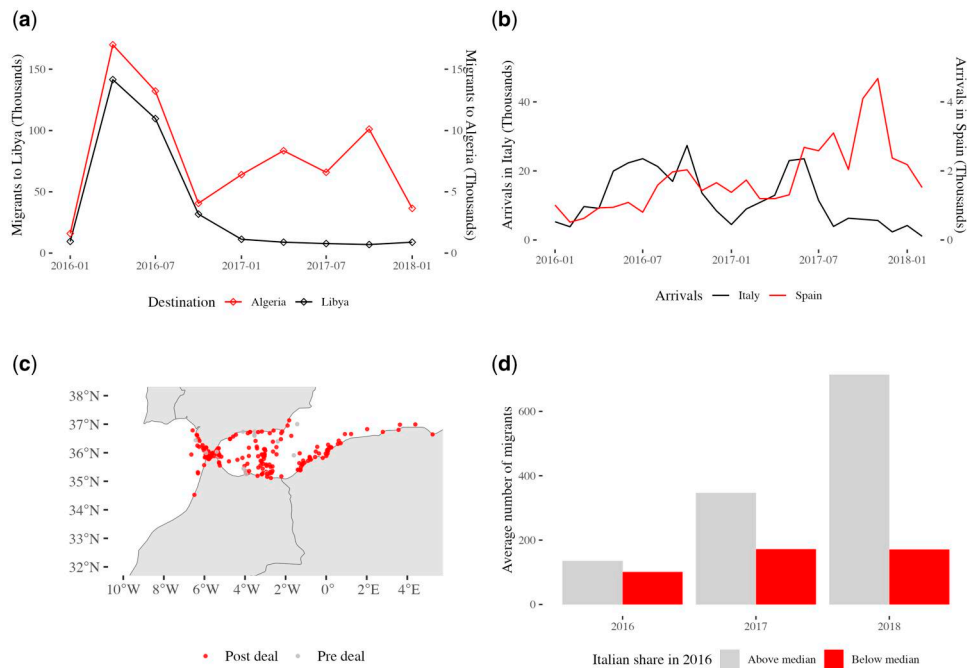


Figure 5. The Western Mediterranean route after the MoU. (a) Migration intentions. (b) Arrivals in Italy and Spain. (c) Deaths in the Western Mediterranean. (d) Arrivals in Spain by country of origin.

Notes: Panel A displays the evolution of migration flows in Niger between 2016 and 2019, measured at two different transit points in the North of the country. The first is Séguedine, the main town on the route to Libya. The second is Arlit, which instead is on the way to Algeria. Panel B displays the number of migrants arriving in Italy and Spain by Sea. The left axis refers to migrants arriving to Italy, the right axis refers to migrants arriving to Spain.

nationalities that take value one if the share of migrants of that nationality was above the median share of migrants arriving to Italy before the signing of the MoU. We then look at the nationality mix of arrivals in Spain and show that migrants from nationalities that were more likely to migrate to Italy are more likely to migrate to Spain after the externalization of SAR activities to the LCG.

Figure 5d shows the average number of migrants arriving in Spain every month of the year. Migrants are divided into two categories: those that were above the median share (gray bars) of nationalities arriving to Italy before the externalization of SAR activities to the LCG and those that were below (red bars). The difference between gray and red bars was relatively low in 2016, before the externalization, and started growing in 2017 and 2018.

We leverage the variation in the nationality mix of arrivals in Spain and estimate the following DD model:

$$y_{nt} = \delta[\text{High Italy}_n \times \text{Post}_t] + \gamma_t + \gamma_n + \varepsilon_{nt}, \quad (3)$$

where y_{nt} represents either the IHS transformation of the number of individuals arriving in Spain in month t from country n or the share of migrants arriving to Spain in month t from country n . γ_t and γ_n represent month and nationality fixed effects. This approach allows us to estimate our coefficient of interest δ , which we can interpret as the average increase in arrivals of individuals from countries that were above the median share of migrants arriving in Italy from a certain country before the signing of the externalization of SAR activities to the LCG. In order to interpret the coefficient δ as a causal parameter, we need to assume that no other shock-affected individuals from countries that we consider as treated, besides the externalization of SAR activities to the LCG.¹⁰

¹⁰ This assumption is relatively strong as it amounts to assume that migration push factors stayed relatively constant in our sample period. In particular, we need to assume that push factors for treated nationalities are not systematically correlated with the timing of the externalization and arrivals in Spain. For example, if individuals from country A were likely to migrate to Italy before the externalization, and individuals from country B were not likely to migrate to Italy, we have to

Table 2. Diversion effect estimates.

	Dependent variable:	
	lhs(Number of Individuals) (1)	Share (2)
High Italy × Post Deal	0.413*** (0.097)	2.020*** (0.756)
Country of Origin	Yes	Yes
Month	Yes	Yes
Observations	696	696
Adjusted R ²	0.937	0.731

The table shows the outcome of two DD regressions in which the outcome is the number of individuals arriving to Spain from country n in month t . The regressor of interest is the interaction between a dummy that takes value 1 that takes value one if the share of migrants of nationality n was above the median share of migrants arriving to Italy before the signing of the MoU and a dummy that takes value 1 for months after January 2017. Standard errors are heteroscedasticity robust. P-values: * $P < .1$; ** $P < .05$; *** $P < .01$.

Table 2 shows the results of this exercise. We estimate that the number of individuals arriving in Spain, who were more likely to migrate to Italy prior to the signing of the MoU, increased by 41 per cent after the signing of the MoU (column (1)). A similar result is obtained for the share of individuals from a specific country arriving to Spain. Results in column (2) show that the share of migrants coming from nationalities that were likely to migrate to Italy prior to the signing of the MoU increased on average by 2 percent.

These figures must be interpreted in light of recent findings on migration cost impacts. Allen, de Castro Dobbin, and Morten (2018) find that higher migration costs, due to the USA–Mexico border wall, prompt migrants to choose different routes rather than deter migration, which is only marginally affected by the border wall. Conversely, Friebe et al. (2024) find that increased costs on the Central Mediterranean route often deter migration rather than affect route choice. Our results align more closely with Friebe et al. (2024), showing that higher costs due to the MoU significantly reduce total arrivals in Europe, with only a fraction shifting to the Western Mediterranean route.

6. Conclusions

Managing large migration flows has proven to be one of the most relevant challenges for policymakers in the most recent decades. The issue has been on the top of the political agenda and has contributed to the boom of extreme far-right parties across Europe and raised the demand for tighter migration policies. The high demand for a reduction in migration flows has caused many countries to sign the so-called *migration deals*, an agreement where destination countries outsource humanitarian assistance and border control to transit or sending countries.

We combine information from several data sources to analyze the effect of outsourcing SAR operations to a country with weak institutions such as Libya. We show that outsourcing SAR operations in this context brought a drastic increase in the probability of observing a deadly event in areas where the LCG was responsible for SAR operations. Our dynamic treatment effect estimates show that the probability of observing a deadly event in areas that were patrolled by the LCG increased when the LCG was equipped with patrol boats to conduct SAR operations, and the Italian MRCC began sharing information about boats in distress with the Libyan authorities. The effect of outsourcing SAR operations on the probability of observing a deadly event becomes close to zero in July 2017. This coincides with three major events: the Italian Government's NGOs code of conduct—which has been shown to decrease the number of migrants departing with less safe boats (Deiana, Maheshri, and Mastrobuoni 2024)—the beginning of monitoring activities by the EU authorities and a sharp drop in departures.

Ultimately, when human lives are at stake it is difficult to derive unilateral policy prescriptions. We show how externalization of SAR activities to politically unstable countries with weak institutions poses a serious risk to the lives of those who need to be rescued. Our findings show that if SAR activities are delegated to an untrained actor the results will be poor. Reassuringly, we show that, in this

assume that there is no migration push factor that after the externalization increases migration from country B more than the increase for country A.

context, the increase in the risk of observing a deadly event closer to the Libyan shores is short-lived. In particular, after a bundle of interventions, which included the monitoring of the activity of the LCG by the EU and the Italian Government's NGO code of conduct, the probability of observing a deadly event in areas where the LCG is more likely to operate SAR activities falls.

Supplementary data

Supplementary data are available at *Journal of Economic Geography* Journal online.

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